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Stream 2 - MID SEMESTER EXAM

Derivative Securities (University of Melbourne)



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THE UNIVERSITY OF MELBOURNE
DEPARTMENT OF FINANCE
MID SEMESTER EXAMINATION—SEMESTER 1, 2023 (STREAM 2)

SUBJECT NUMBER:	FNCE30007
SUBJECT NAME:	Derivative Securities
EXAMINATION DURATION:	1 hour
READING TIME:	5 minutes
TOTAL MARKS:	12
NO. OF PAGES:	10

Candidate's Enrolment Number:
Seat Number:

(Do NOT write your name anywhere on this exam booklet)

AUTHORISED MATERIALS

1. This is a CLOSED BOOK examination.
2. Only the approved calculator – Casio FX82 (any suffix) may be used. No equivalent models are permitted.
3. Students are NOT permitted to use any dictionary.

INSTRUCTIONS TO INVIGILATORS

1. Students should be provided with this booklet and a multiple choice answer sheet.
2. Writing is NOT permitted during reading time.
3. This booklet and the multiple choice answer sheet must be COLLECTED at the end of the exam.

INSTRUCTIONS TO STUDENTS

1. Writing is NOT permitted during reading time.
2. Write your seat number and student number in the boxes at the top of this cover page. Also complete the following details on the multiple choice answer sheet.
   - Write Subject Code
   - Write your Surname
   - Write your Student Number and shade the corresponding oval beneath each number. **Ensure the correct oval is shaded**.
3. Answer ALL questions. No marks will be deducted for incorrect responses.
4. 3 pages are provided for rough work.
5. A formula list is provided on the last page. You may detach these pages.
6. Do NOT detach any other pages.
7. Do NOT round intermediate calculations, rounding should ONLY be performed on the **final answer**.
8. All questions are worth 1 mark each.
9. **This paper must be handed in at the end of the examination. Do NOT remove this paper from the examination room.**

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Question 1

Consider an equity portfolio with a beta of 0.8 currently valued at $10,000,000. The fund manager seeks to increase the risk of the fund over the next 2 weeks by increasing the beta to 1.35. The index futures are trading at 5,500 points. Each contract point is worth $15. The fund manager should (to the nearest integer)

(a) Long 67 contracts
(b) Long 24 contracts
(c) Short 24 contracts
(d) Short 67 contracts
(e) None of the above

Question 2

Which of the following is NOT true? The minimum variance hedge ratio

(a) Is optimal when the hedge is closed out prior to the maturity date of the futures contract because the basis risk is not zero.
(b) Should be implemented when cross hedging
(c) Is only equal to unity (one) when the correlation between the spot and the futures is unity (one).
(d) Does not always provide a greater level of risk reduction than a naïve hedge
(e) None of the above

Question 3

Consider a speculator seeking to trade volatility (standard deviation) on the S&P500 index. At the money put options on the S&P500 with a maturity date of 4 months are currently priced using the market consensus forecast of 25% p.a. The speculator has an econometric model that forecasts volatility to be 28% p.a over the next 4 months. The best trading strategy for the speculator is

(a) Short deep out of the money puts and close out the position in 4 months' time.
(b) Long deep out of the money calls and close out the position in 4 months' time.
(c) Long at the money puts and close out once the higher volatility expectations are reflected in the option price.
(d) Short at the money puts and close out as soon as the higher volatility expectations are reflected in the option price.
(e) None of the above.

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Question 4

Suppose the one-year interest rates in Australia and Canada are respectively 3.5% p.a and 5% p.a continuously compounded. The spot exchange rate is 1AUD = 1.1154CAD. The one year forward rate (to 4 decimal places) is

(a) 1AUD = 1.0988 CAD
(b) 1AUD = 1.0621 CAD
(c) 1AUD = 1.1277 CAD
(d) 1AUD = 1.1323 CAD
(e) None of the above.

Question 5

The futures price of a consumption commodity cannot be solely determined by the underlying spot price, interest rate, storage costs and time to maturity because

(a) Convenience yields are unobservable
(b) Arbitrageurs are unable to take a long position in the underlying asset
(c) The transaction costs associated with arbitrage are too high.
(d) Arbitrageurs are unable to take a short position in the underlying asset
(e) None of the above.

Question 6

If there is a strong negative correlation between interest rates and the spot price, then holding everything else equal

(a) Futures are priced lower than forwards because agents with short futures positions can finance their losses at a higher interest rate
(b) Futures are priced higher than forwards because agents with a long futures position can invest their profits at a lower interest rate
(c) Futures are priced higher than forwards because they have favourable tax treatment.
(d) Futures are priced lower than forwards because agents with short futures positions can invest their gains at a higher interest rate.
(e) None of the above.

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Question 7

Consider a risk averse food producer at March 20, 2023 seeking to hedge corn purchases from March 21, 2023 to March 23, 2023. The producer employs corn futures contracts that expire March 31, 2023. An analysis of daily historical data reveals the following

Correlation between corn spot and corn futures returns = 0.92
Variance of corn spot returns = 2.2
Variance of corn futures returns = 2.5

The following daily return data is observed over the life of the hedge

Date	Spot return (%)	Futures return (%)
March 21, 2016	1.50	1.65
March 22, 2016	-0.80	-0.70
March 23, 2016	2.00	1.90

To calculate variances use the formula for a population: $Var(X) = \frac{\sum_{i=1}^{N}(X_i - u)^2}{N}$. You are not required to tail the hedge

To 4 decimal places, which of the following statements is true?

(a) The hedged returns over the 3-day period are 0.2403% with a variance of 0.0516%.
(b) The hedged returns over the 3-day period are 0.1500% with a variance of 0.0117%.
(c) The hedged returns over the 3-day period are -0.2403% with a variance of 0.0516%.
(d) The hedged returns over the 3-day period are -0.1500% with a variance of 0.0117%.
(e) None of the above

Question 8

Consider a forward contract for a coupon bond. The spot price for the bond is currently $950. The forward contract has two years to maturity and coupon payments of $100 are due in 6 months and 18 months' time. Continuously compounded risk-free interest rates % p.a are

Period	Rate (%)
1 month	3.50
6 months	4.00
12 months	4.50
18 months	5.00
24 months	6.00

The forward price to 2 decimal places should be

(a) $858.66
(b) $865.02
(c) $971.41
(d) $855.54
(e) None of the above

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Question 9

Consider a European call option currently priced at $2.50 on a dividend paying stock priced at $22. The call has a strike of $15 and a maturity of 6 months. A dividend of $4.0605 is expected in 3 months. Assuming a risk-free rate of 6% p.a continuously compounded and a flat term structure of interest rates, which of the following statements is true to 4 decimal places?

(a) An arbitrageur could buy the call, short the stock and sell bonds, realising an immediate profit of $0.9433.

(b) An arbitrageur could sell the call, buy the stock and sell bonds, realising an immediate profit of $0.9433.

(c) An arbitrageur could buy the call, short the stock and buy bonds. The maximum profit from the arbitrage in today's dollars is $0.9433.

(d) An arbitrageur could buy the call, short the stock and buy bonds. The minimum profit from the arbitrage in today's dollars is $0.9433.

(e) None of the above

Question 10

Consider an American put option on a non-dividend paying stock. The stock is currently priced at $60 and the put is priced at $3. The option has 6 months to expiration and a strike of $66. The risk-free rate is 8% p.a. continuously compounded and the market expects it to rise to 9% p.a over the coming week. Which of the following statements is true?

(a) An American put should never be exercised early.

(b) An arbitrageur can buy the put and the stock, and exercise immediately, earning a profit of $3.

(c) An arbitrageur can buy the put and the stock and exercise immediately, earning a profit of $2.8824.

(d) An arbitrageur can sell the put and stock and buy bonds, immediately earning a profit of $2.8824.

(e) None of the above.

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Question 11

Consider a European call and put option on an underlying stock currently trading at $25. Both options have a maturity of 3 months and a strike of $23. The put is priced at $1.50 and the call is trading at $6. The risk-free rate is 5% p.a continuously compounded. Which of the following statements is true to 4 decimal places?

(a) An arbitrageur can sell the call and bonds, buy a put and the stock, and immediately receive a riskless profit of $2.5143. Irrespective of the stock price at maturity, the arbitrageur has locked in a buying price of $23 for the stock at maturity.

(b) An arbitrageur can sell the call and bonds, buy a put and the stock, and immediately receive a riskless profit of $2.2143. Irrespective of the stock price at maturity, the arbitrageur has locked in a buying price of $23 for the stock at maturity.

(c) An arbitrageur can buy the call and bonds, sell a put and the stock, and immediately receive a riskless profit of $2.2143. Irrespective of the stock price at maturity, the arbitrageur has locked in a buying price of $23 for the stock at maturity.

(d) An arbitrageur can sell the call and bonds, buy a put and the stock, and immediately receive a riskless profit of $2.2143. Irrespective of the stock price at maturity the arbitrageur has locked in a selling price of $23 for the stock at maturity.

(e) None of the above

Question 12

Consider a speculator who has a bearish view on the ASX200. The speculator seeks to use $500,000 to take the appropriate position in SPI200 futures which are currently traded at 5,800 points. Each SPI futures contract requires a $20,000 margin and each point is worth $25. Assume that the SPI200 futures contract increases by 6% and the speculator closes out their position. The speculators total profit or loss is

(a) $8,700 loss

(b) $217,500 profit

(c) $8,700 gain

(d) $219,500 loss

(e) None of the above

End of Examination

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SPACE FOR ROUGH WORK
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FORMULAE SHEET

$F_0 = S_0 e^{rT}$

$F_0 = (S_0 - I)e^{rT}$

$F_0 = S_0 e^{(r-q)T}$

$F_0 = S_0 e^{(r-r_f)T}$

$F_0 \le (S_0 + U)e^{rT}$

$F_0 \le S_0 e^{(r+u)T}$

$F_0 e^{yT} = (S_0 + U)e^{rT}$

$F_0 = S_0 e^{(r+u-y)T}$

$b_t = S_t - F_t$

$r_i = r_f + \beta(r_m - r_f)$

$h^* = \frac{\rho_{sf}\sigma_s}{\sigma_f} = \frac{\sigma_{sf}}{\sigma_f^2}$

$N^* = \frac{h^* Q_A}{Q_F}$

$N^* = \beta \frac{P}{F}$

$(\beta - \beta^*)\frac{P}{F}$

$(\beta^* - \beta)\frac{P}{F}$

$F_0 + B_1$

$F_0 + (S_1^* - F_1) + (S_1 - S_1^*)$

$f = (F_0 - K)e^{-rT}$

$f = (K - F_0)e^{-rT}$

$c \ge \max(S_0 - Ke^{-rT}, 0)$

$p \ge \max(Ke^{-rT} - S_0, 0)$

$c + Ke^{-rT} = p + S_0$

$S_0 - K \le C - P \le S_0 - Ke^{-rT}$

$c \ge \max(S_0 - D - Ke^{-rT}, 0)$

$p \ge \max(D + Ke^{-rT} - S_0, 0)$

$c + D + Ke^{-rT} = p + S_0$

$S_0 - D - K \le C - P \le S_0 - Ke^{-rT}$

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